

C.O.S.M.O

COGNITIVE OPERATIONS & SECURITY MONITORING ORCHESTRATOR

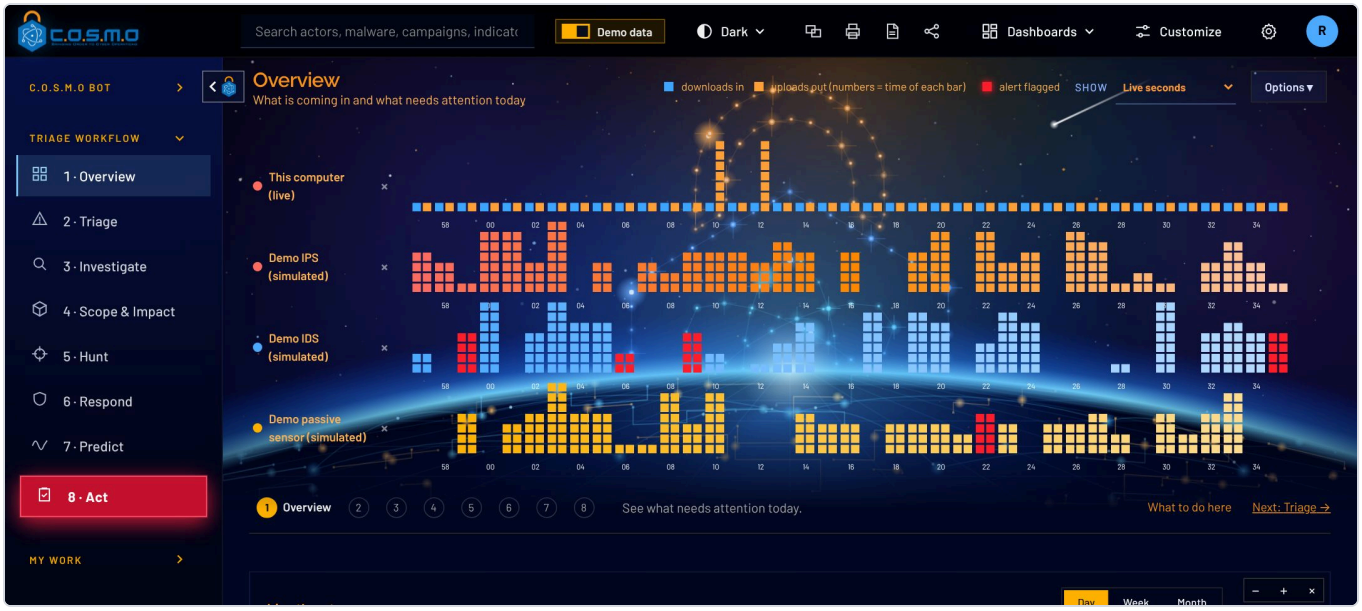
Bringing order to cyber operations.

Threat intelligence, triage, investigation and response in one platform that runs on your own systems, with the AI you choose, or none at all.

WHY C.O.S.M.O

C.O.S.M.O turns a flood of alerts into decisions your team can stand behind.

Security teams drown in alerts, scattered intelligence and tools that do not talk to each other. C.O.S.M.O brings them into one place: it watches your network, compares what it sees with what the world knows about attackers, and guides your analysts from "something happened" to "it is handled, and we know who did what."



The Overview Live activity from every source, the threat map, and a guide that tells each analyst what to do next.

It watches

Firewall, intrusion detection and server logs, cloud buckets, databases and your SIEM. Every address, domain and file fingerprint is checked against known threats.

It knows

MITRE ATT&CK, CISA's exploited-vulnerability catalog, ransomware leak sites, research blogs and your own feeds, kept as standard STIX 2.1.

It guides

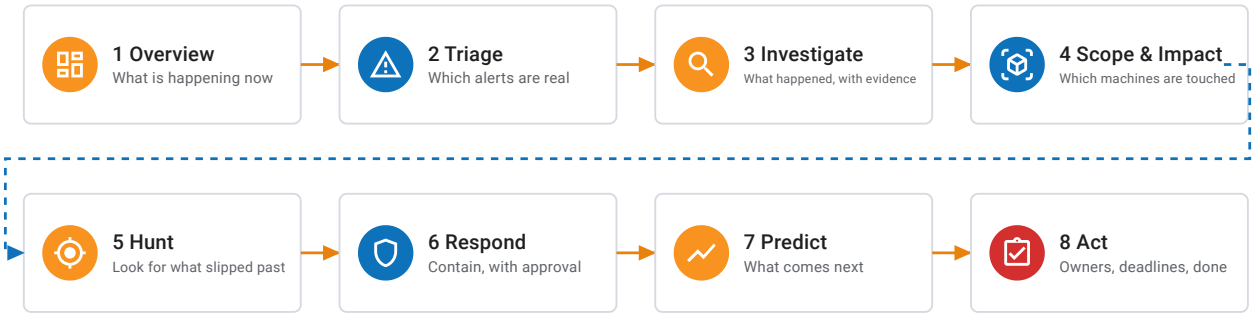
Every page explains itself in plain language, and C.O.S.M.O Bot answers questions about your own data using the AI you connect.

8	26	20+	0
GUIDED WORKFLOW STEPS	INTELLIGENCE SOURCES BUILT IN	LOG FORMATS READ	CLOUD ACCOUNTS REQUIRED

ONE WORKFLOW

Eight plain steps take every incident from first alert to finished work.

Analysts never wonder what to do next. Each step answers one question and hands its result to the next, following the NIST incident-handling cycle with intelligence work on both sides.



Step	The question it answers	What your team leaves with
1 · Overview	Is anything unusual happening right now?	A short list of things worth a look
2 · Triage	Which of these alerts are real, and which matter most?	Every alert decided, with a reason
3 · Investigate	What actually happened, and how do we know?	A case with evidence and a conclusion
4 · Scope & Impact	Which of our machines are involved?	The affected hosts, weighted by importance
5 · Hunt	Is the same attacker anywhere else?	Hunts across all your data, findings filed as cases
6 · Respond	What do we block, isolate or reset?	Actions a person has approved
7 · Predict	What is this attacker likely to do next?	Predictions with evidence and a checklist
8 · Act	Who does what, by when?	A playbook where every action has an owner and a due time

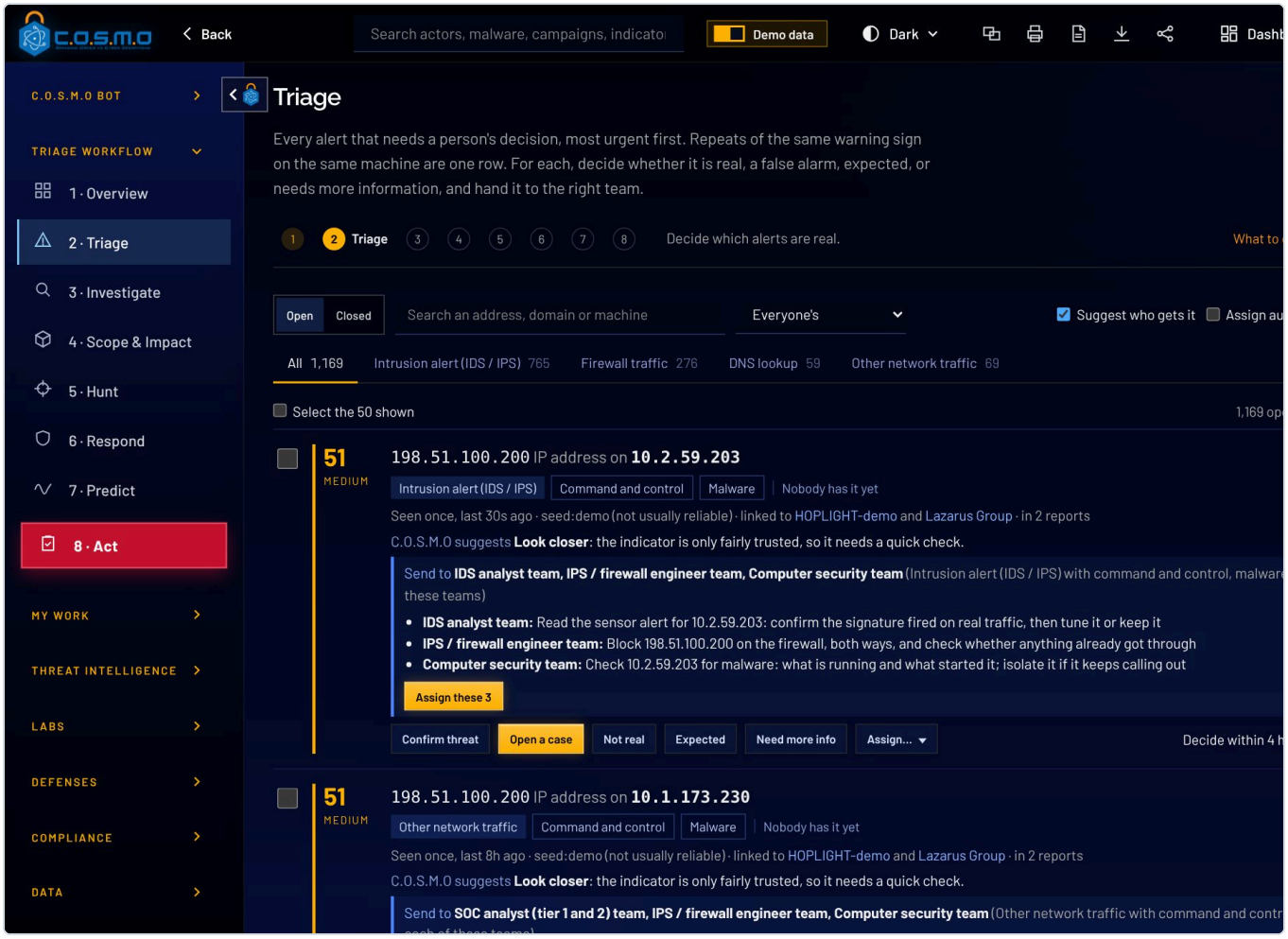
Built for analysts at every level

A new analyst follows the guide on each page; an experienced one jumps straight to the step they need. Both leave the same clear record.

TRIAGE

Triage separates what is real from what is noise, and shows why.

Repeated alerts are grouped into incidents, each with a priority score your analysts can read and question. Decisions teach C.O.S.M.O what your environment considers normal.



Step 2 · Triage Incidents ranked by a score that explains itself, with service-level timers and one-click decisions.

Faster decisions

- Repeats grouped into one incident
- A score that lists its reasons
- Service-level timers per priority
- Real, benign, false positive or needs more information

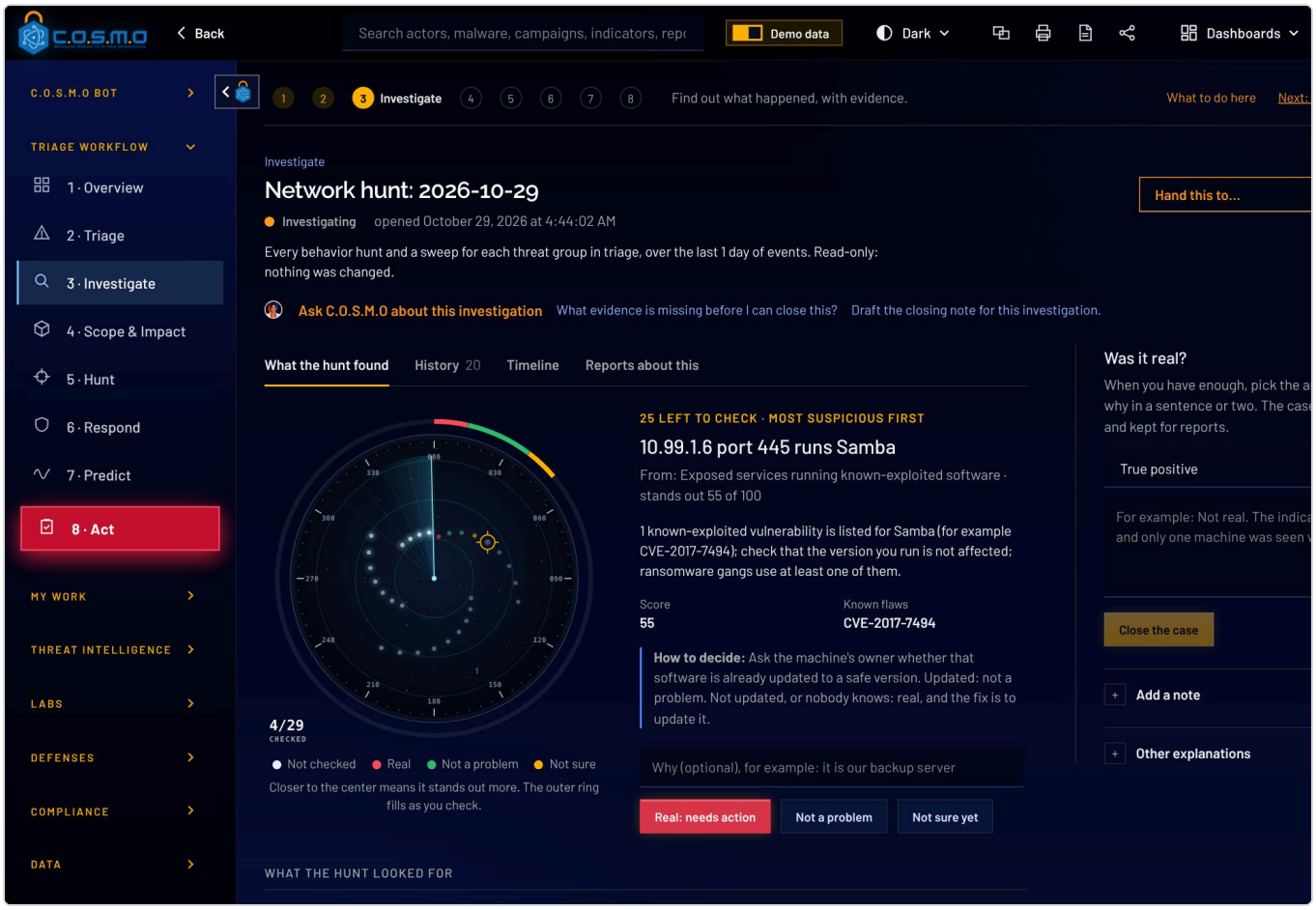
Smarter over time

- Sources graded A to F for reliability
- Machines weighted by how much they matter
- Past decisions inform new ones
- Every decision recorded with who and when

INVESTIGATE AND HUNT

Every investigation keeps the evidence, the reasoning and the people in one place.

When something is real, a case holds the questions, the evidence and the conclusion. Scope shows which machines are touched, and hunting finds the same attacker wherever else they are.



Step 3 · Investigate A guided case: what to check, what was found, and a conclusion the whole team can read.

Investigate

Guided questions, evidence from your own logs and SIEM, and a plain-language opinion that stays consistent from analyst to analyst.

Scope & Impact

The affected hosts on a map of your network, weighted by how critical each one is, with the data quality behind every answer.

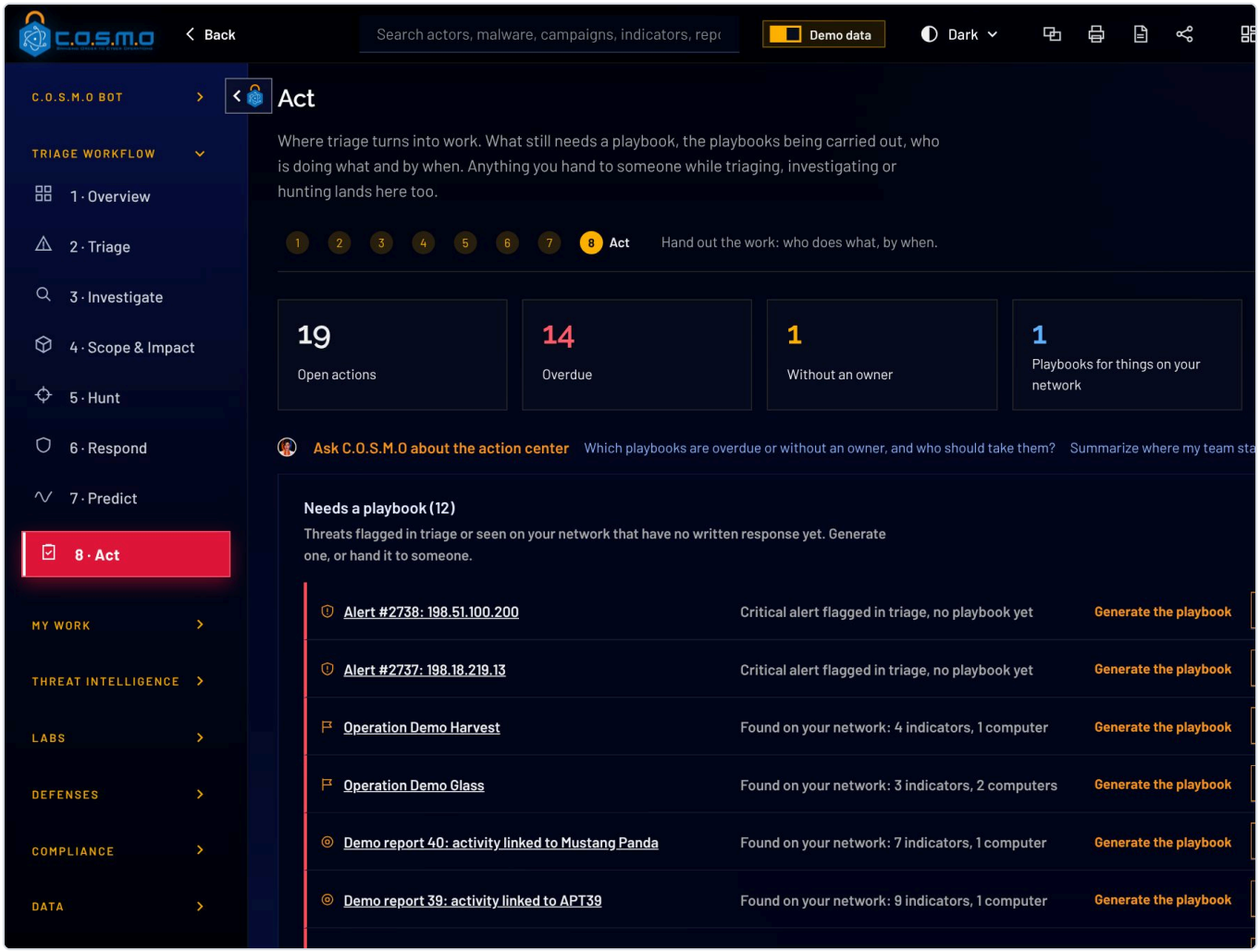
Hunt

ATT&CK-mapped hunts and Sigma, YARA, Suricata and Snort rules run across all your data, plus beaconing, DNS tunneling and generated-domain detection.

RESPOND AND ACT

People stay in charge of every response, and every action has an owner.

C.O.S.M.O proposes the block, the isolation or the password reset. A person approves it. Then the work becomes a playbook with owners and deadlines, so nothing falls between teams.



Step 8 · Act One board for the whole team: what needs a playbook, what is in progress, and who is doing what.

Respond

Containment proposed from the evidence and approved by a person. AI and integrations can suggest; only a signed-in person can approve.

Predict

What this attacker is likely to try next, based on their known techniques, with a checklist to get ahead of it.

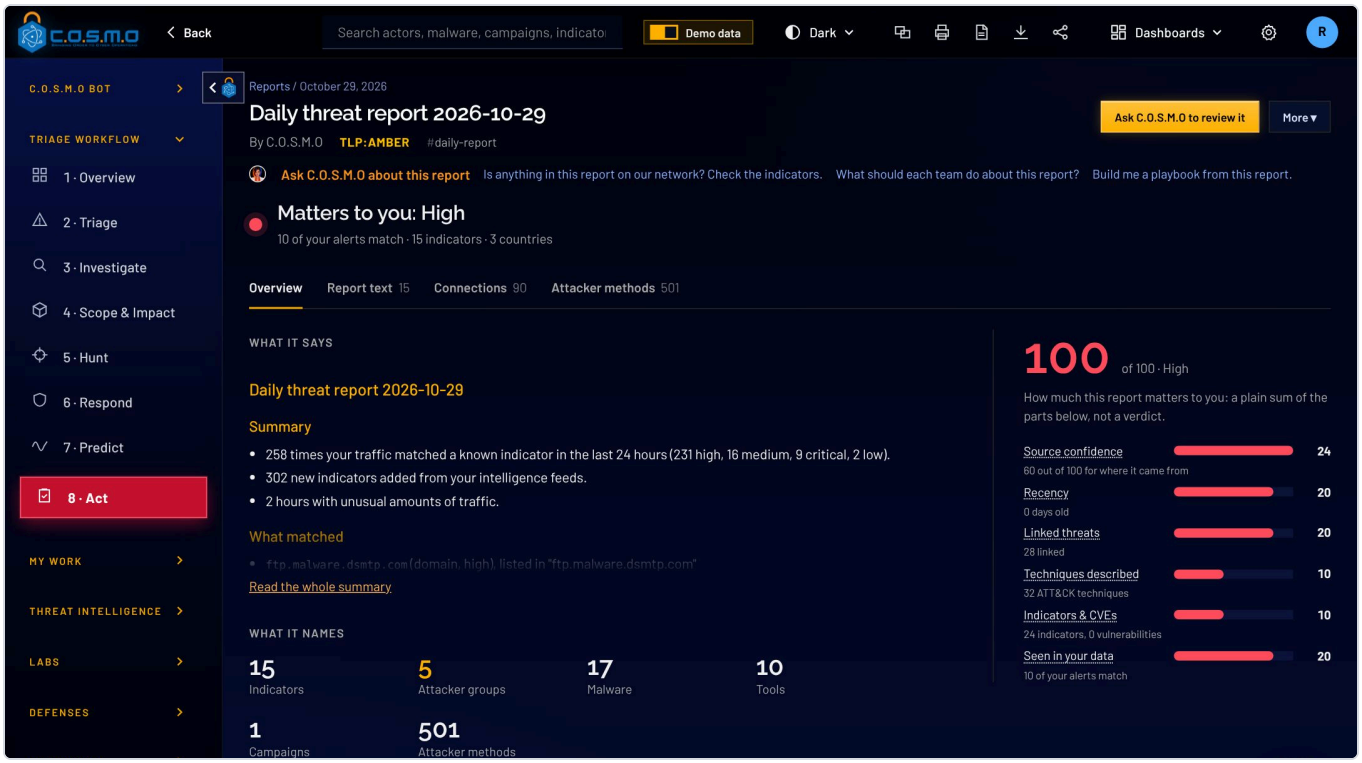
Act

Playbooks where every action has a phase, a duty, an owner and a due time, assigned from any step and tracked to done.

THREAT INTELLIGENCE

Threat intelligence arrives on day one and keeps itself current.

C.O.S.M.O collects open-source intelligence by itself on first start and refreshes it on schedule. Reports become structured knowledge you can explore, rank and tie to your own alerts.



Reports Each report broken into actors, malware, techniques and indicators, with a relationship graph and an ATT&CK timeline.

A knowledge base, not a pile of PDFs

- Standard STIX 2.1 throughout
- Threat Library of actors, malware, campaigns, tools and vulnerabilities
- 2D and 3D relationship graphs
- Most-wanted and dark-web monitoring

Sources you already trust

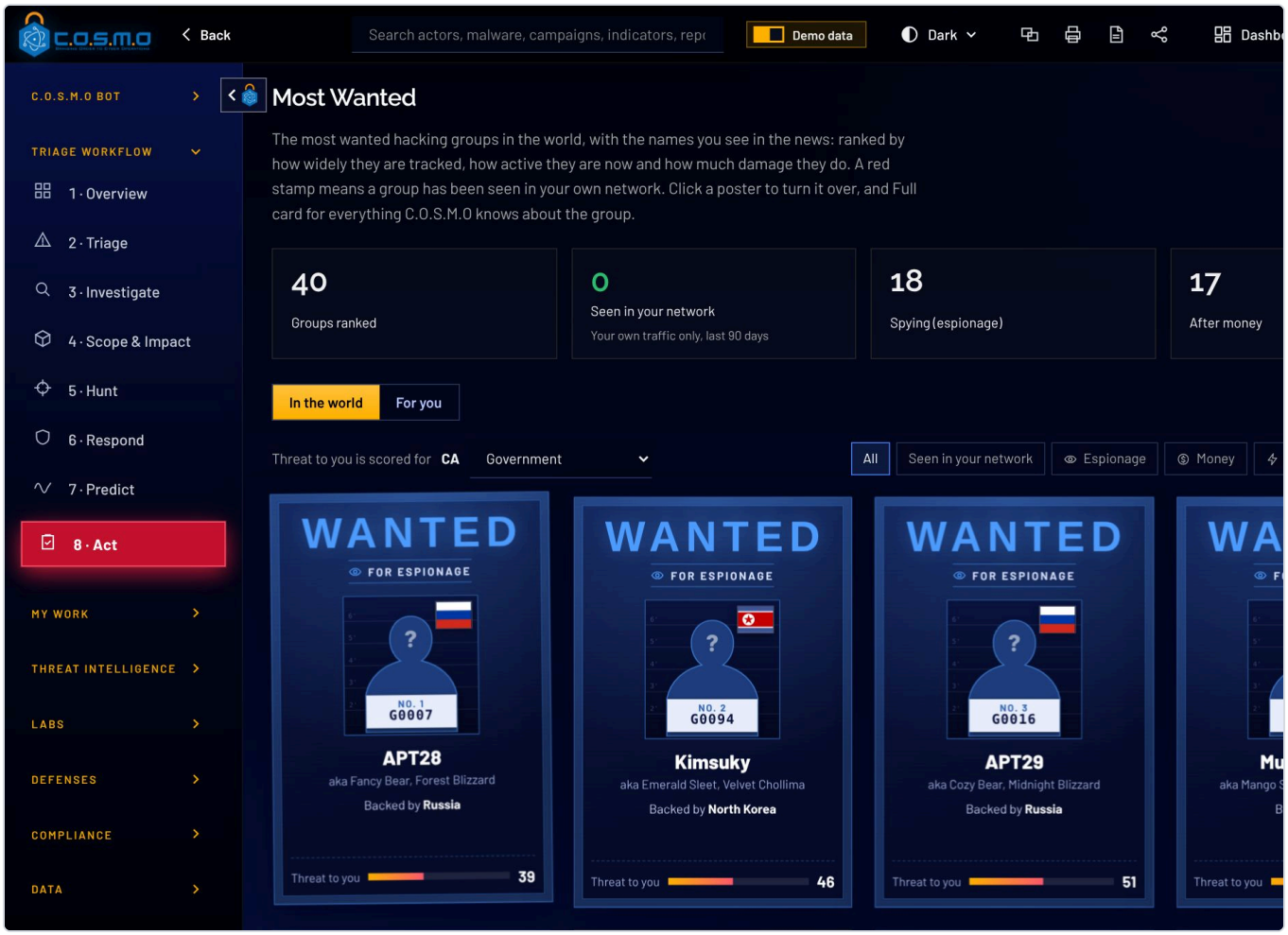
- MITRE ATT&CK, CISA KEV, abuse.ch
- MISP, OTX and any TAXII 2.1 server
- Research blogs and ransomware leak sites
- Closed networks load intelligence from files

MITRE	CISA	MISP	OTX	TAXII	
ATT&CK	KEV	Events	AlienVault	Any 2.1 server	Hash look-ups

MOST WANTED

Know your adversaries by name, before they know you.

The world's most dangerous hacking groups as wanted posters, under the names you see in the news. **In the world** ranks them by reach and activity; **For you** ranks the same groups for your sector, and any group seen in your own network goes to the top.



Most Wanted Each poster's color is its motive: blue espionage, gold money, red sabotage, violet ideology. A red stamp means the group has been seen in your network.

Read it in seconds

Who they are, who backs them, why they attack and how big a threat they are to you, on one poster.

Turn it over

How they work step by step, their tools and techniques, the reports about them, and where they were seen in your data.

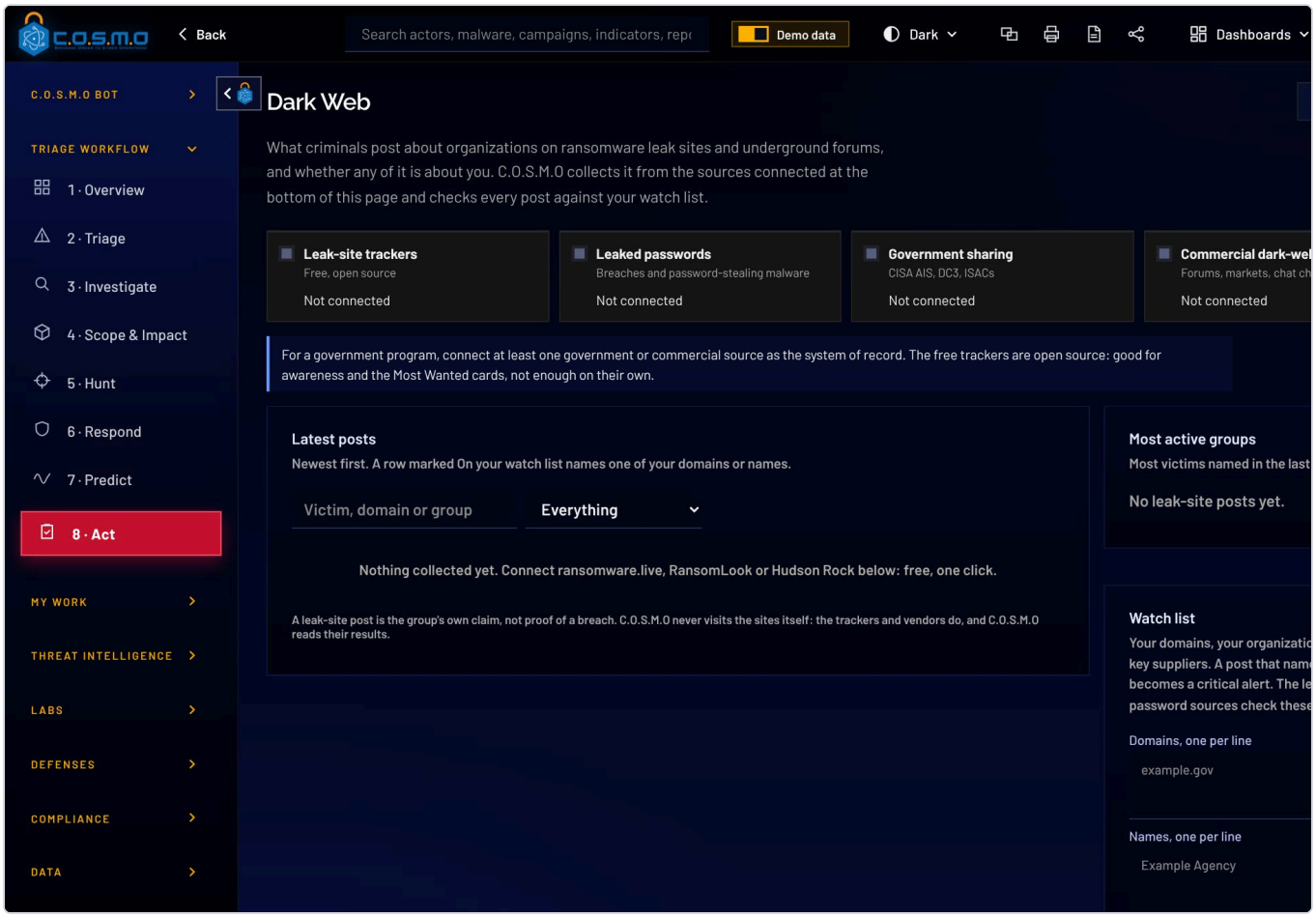
Hunt them now

One click sweeps all your data for every indicator and technique tied to that group, and files what it finds as a case.

DARK WEB

Find out when criminals name you, before your customers do.

Ransomware gangs name their victims on leak sites, and criminals sell stolen logins on underground forums. C.O.S.M.O reads those sources for you and checks every post against your **watch list**: your domains and your organization's names.



Dark Web Latest leak-site posts, the most active groups and your watch list. A match turns red and goes straight to Triage. Shown with sample data.

Leak sites

Victims posted by ransomware groups, from free trackers or the government and commercial sources you already use.

Stolen passwords

Computers infected by password-stealing malware that signed in to your domains, and your accounts in known breaches.

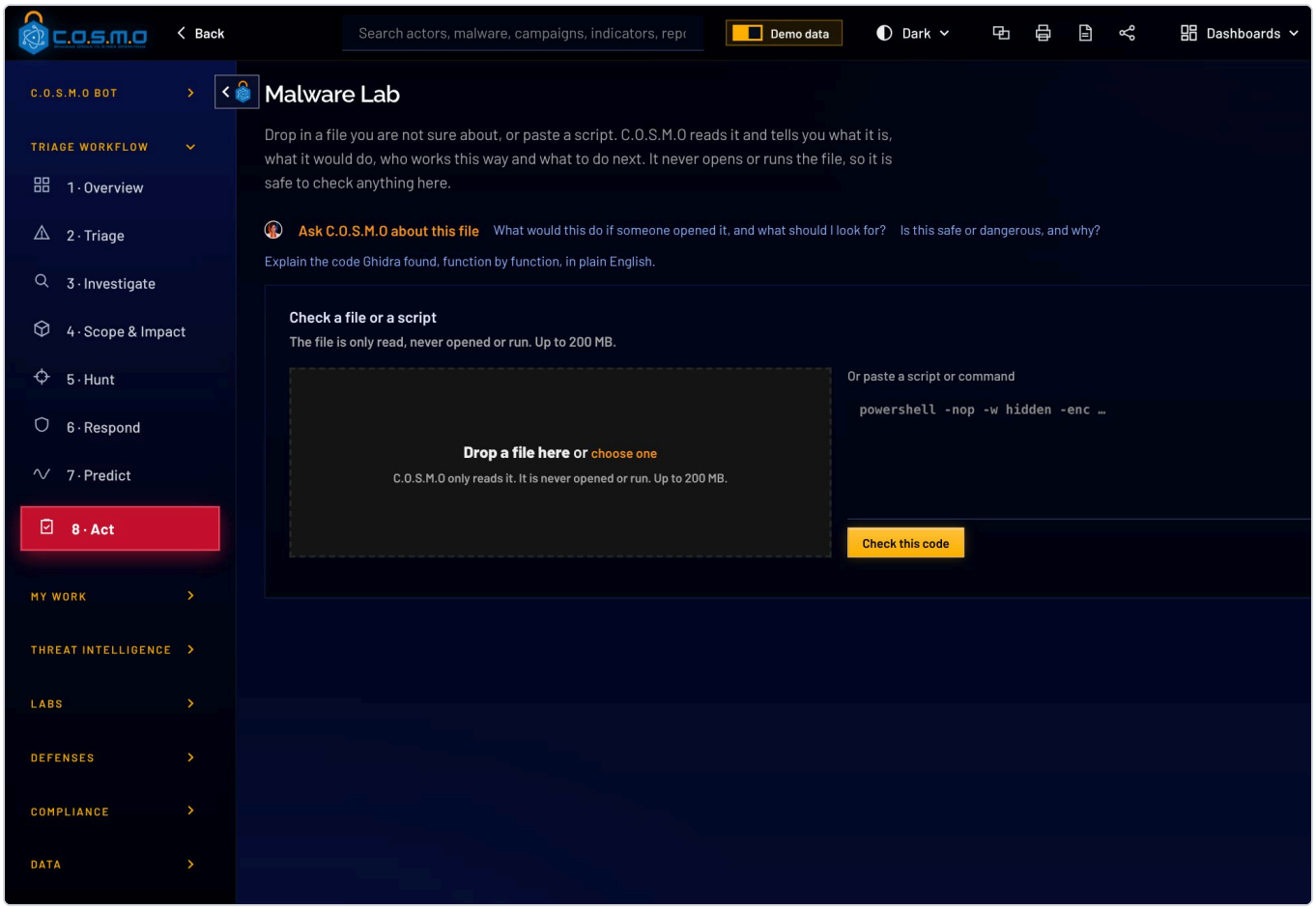
Straight to Triage

A match becomes a critical incident in Triage, with a one-click hunt to check whether the leaked logins are being used.

LABS

Suspicious files are taken apart safely, down to the code.

Drop a file, a script or a compiled program into the **Malware Lab**. C.O.S.M.O reads it without ever running it, and for a program it also takes the code apart with **Ghidra**, the NSA's free reverse-engineering tool, to show what the program can do.



Malware Lab Drop a file or paste a script. It is read, never run, and comes back as a one-page report.

A plain report

How dangerous it is, what it would do, where it would connect, who works this way, and what to do next.

The code, explained

What a program can do, each ability mapped to MITRE ATT&CK, signs of packing, and the code of its riskiest functions. C.O.S.M.O Bot explains it line by line.

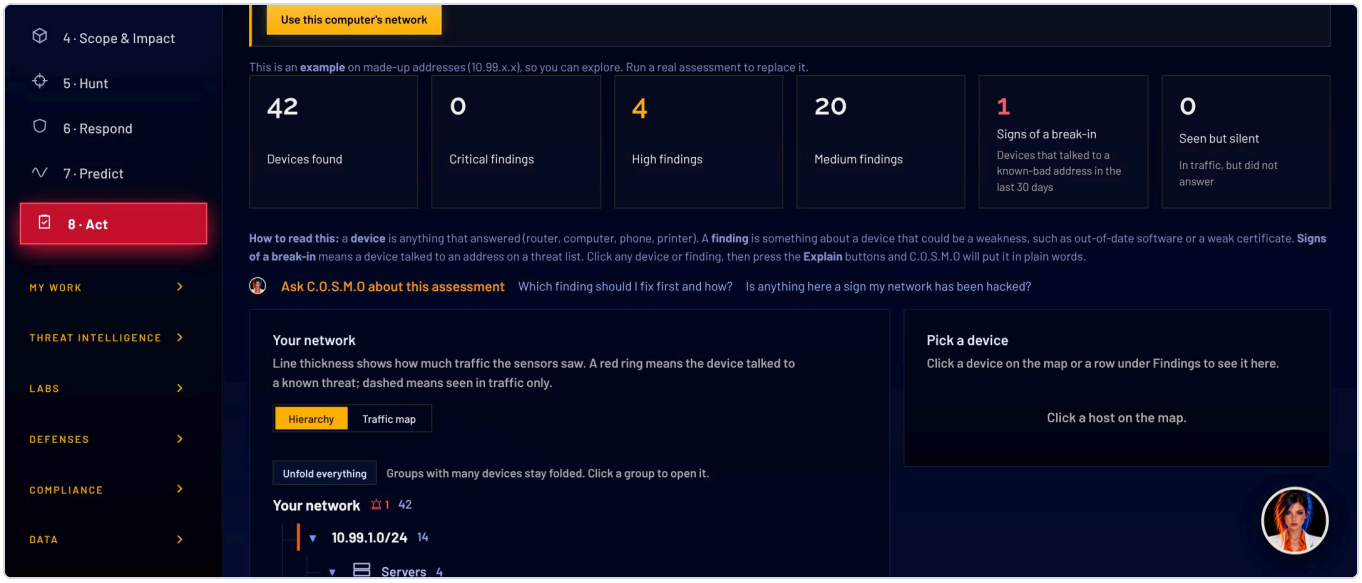
Forensics too

Drop a collection from a computer and get one timeline of what happened, with what stands out flagged and tied to known groups.

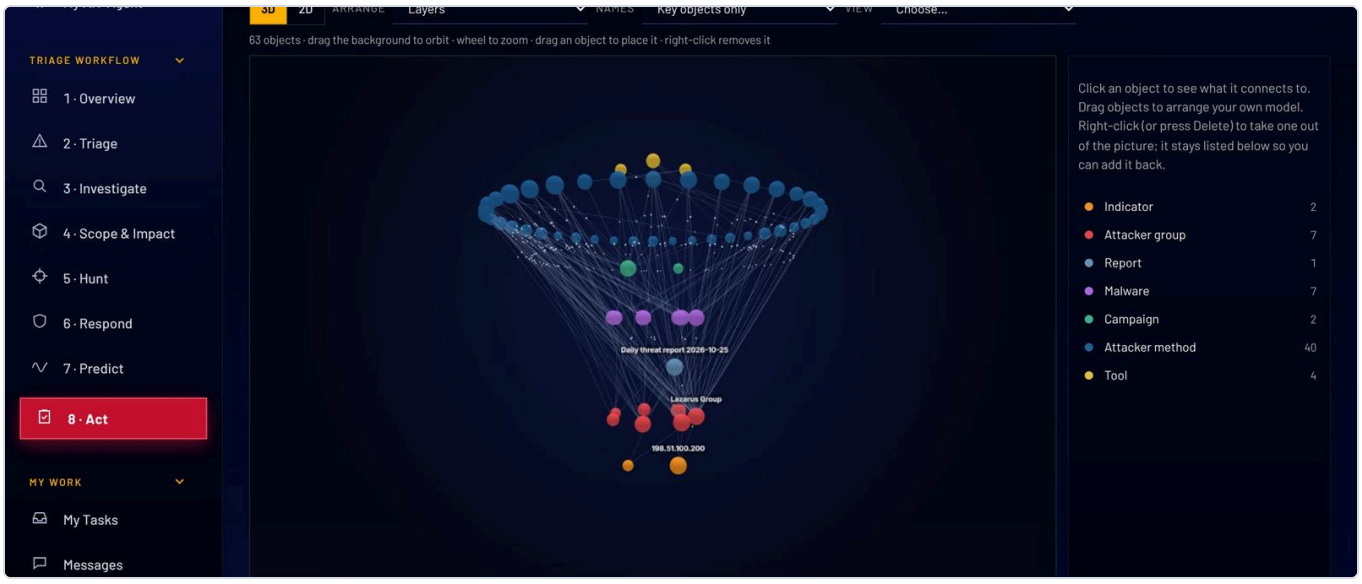
SEE IT

Complex threats become pictures your whole team understands.

Network discovery draws what is actually on your network and what is exposed. The knowledge graph shows how an attacker, their malware, their techniques and your alerts connect, in 2D or 3D.



Network Machines found, open and weak services, and the hosts that talked to known threats, drawn as a map you can fold and unfold.

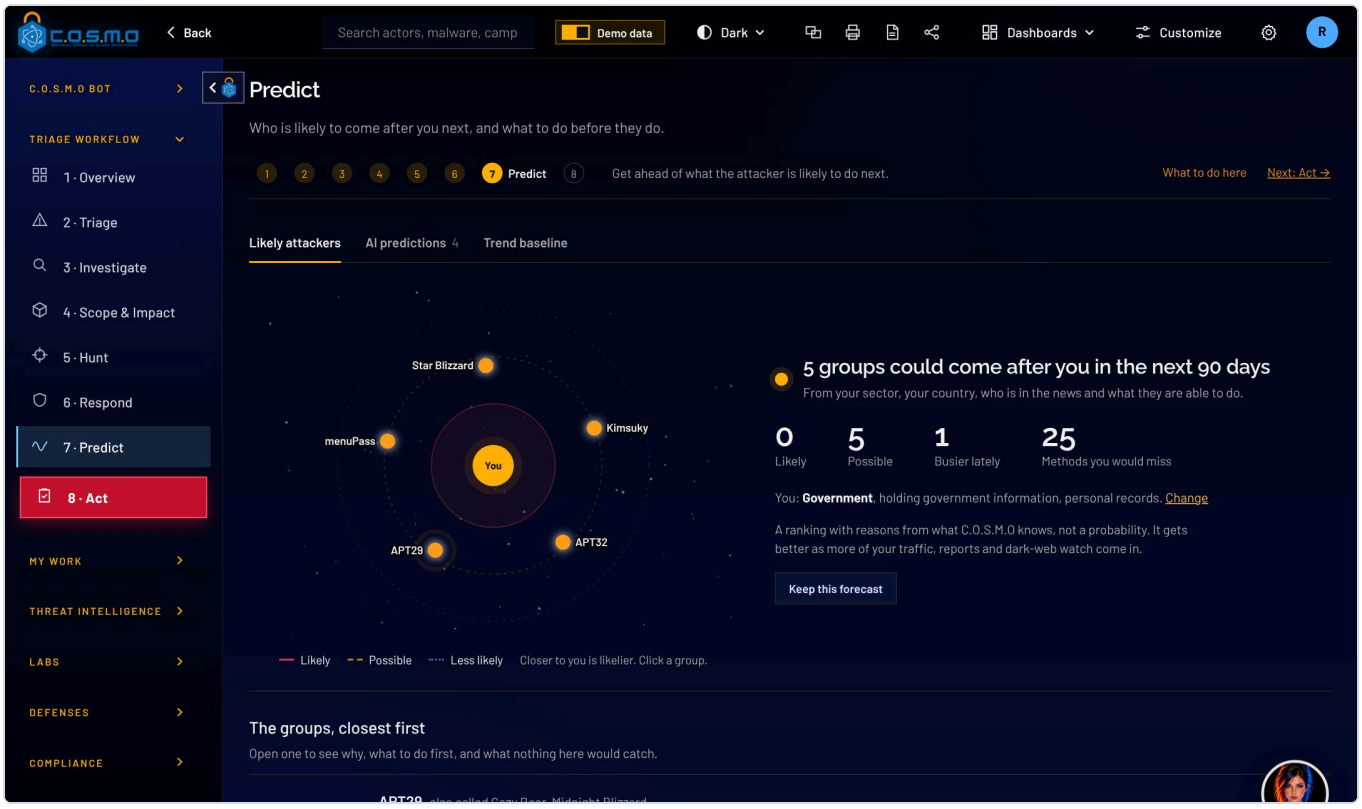


Knowledge graph Every object in a report and how it connects. Rotate it, zoom in, drag objects into place, and click one to see what it touches.

LOOK AHEAD

Get ahead of attackers instead of only reacting to them.

C.O.S.M.O names the groups most likely to come after you next, shows how AI services are really used on your network, and checks the suppliers and services your organization cannot do without.



Who is coming next The groups most likely to target you in the next 90 days, why, whether they are getting busier, and what to do first.

Who is coming next

A ranked forecast with its reasons, kept every week and scored against what really turned up 90 days later.

AI use on your network

Which machines use which AI services, unapproved "shadow AI", large uploads and servers calling models. Never what anyone typed.

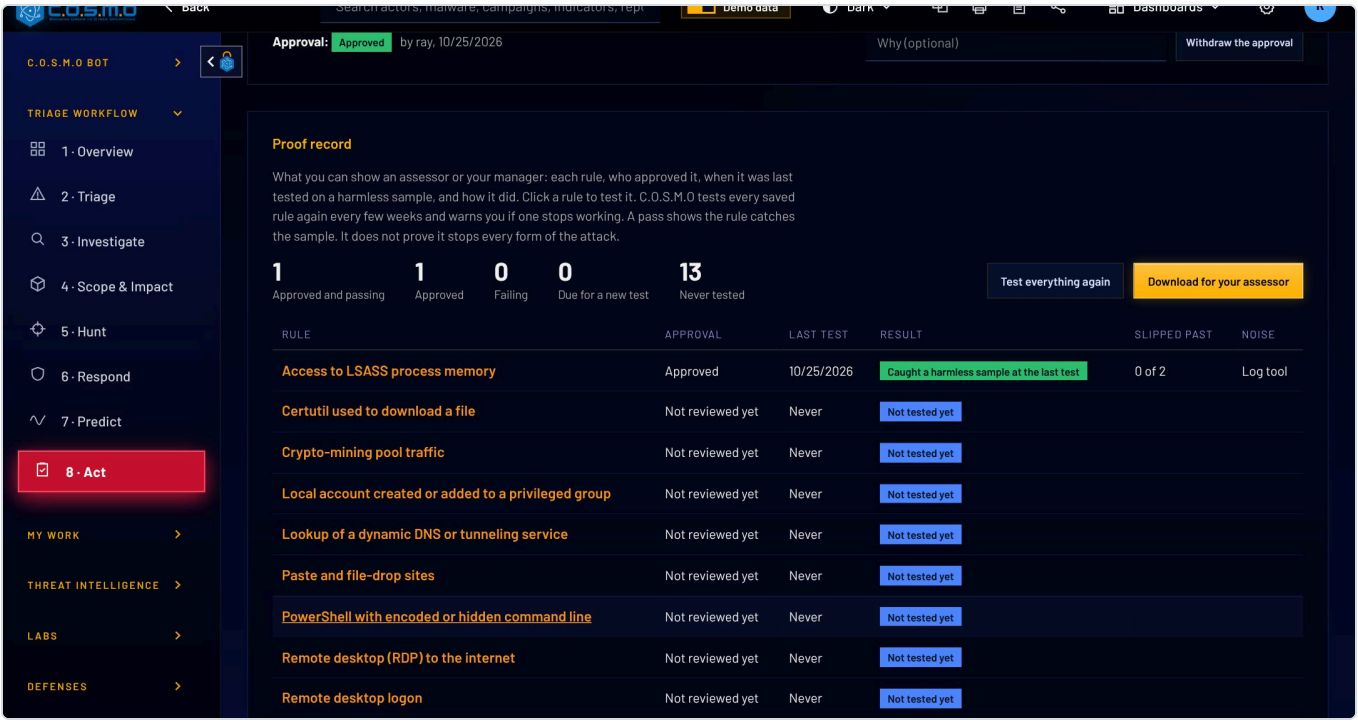
Your suppliers, services and people

Suppliers checked against leak sites, exploited flaws and leaked logins. Alerts say which service a machine runs, what depends on it, and whose machine it is. Your directory (Active Directory, Entra ID, Okta) shows the admin accounts attackers like.

PROOF, NOT PROMISES

Show your assessor, your board and your insurer what really happened.

Detection rules are proven with harmless samples and approved by a person. Every incident keeps a timeline of decisions anyone can check was not edited. Every piece of intelligence says how sure it is.



Proof of Defense Every rule with its last test, what slipped past, its noise on real traffic and who approved it.

Proof of Defense

- Each rule tested with a harmless sample
- Saved as proof and approved by a person
- Re-tested on its own; a failure raises an alert
- One download for your assessor

Incident timeline and record

- Alert, decisions, approvals and actions in order
- Other explanations weighed and recorded
- A record with a fingerprint in the sealed log
- Paste a copy back to check it was not edited

How sure are we

Every source behind a threat, how far it is trusted and where sources disagree, summed up in the code analysts already use (for example B2).

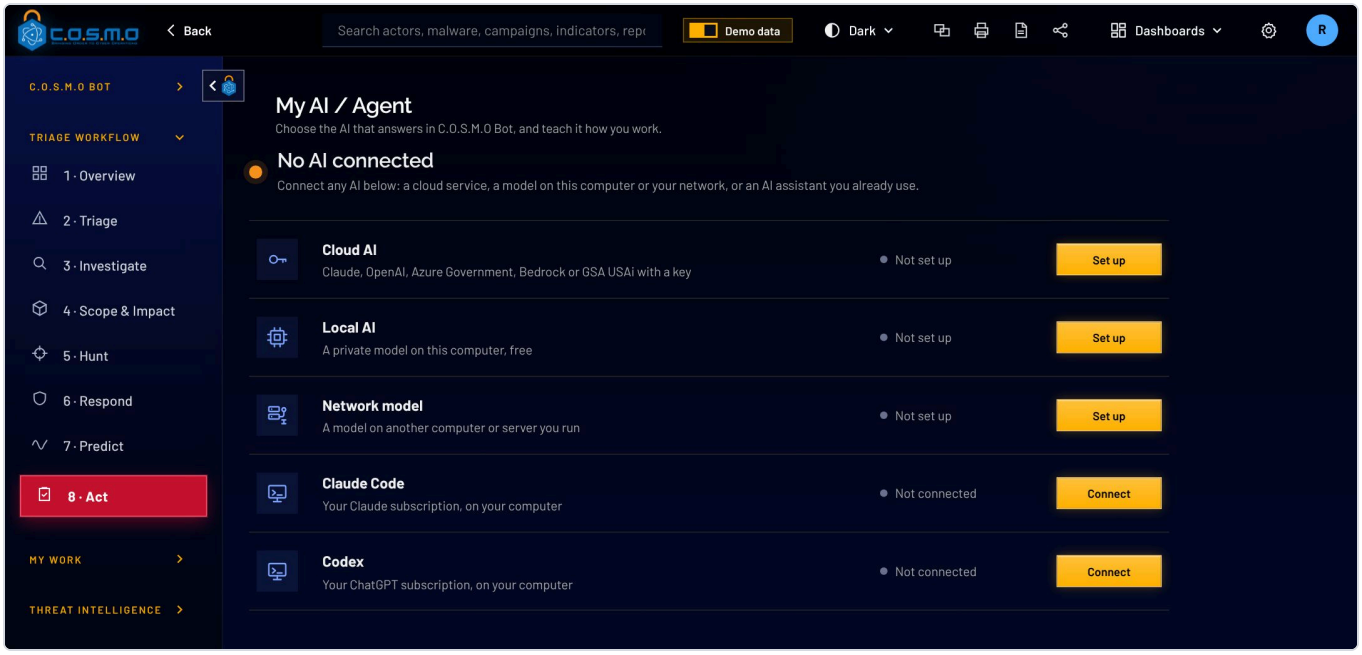
C.O.S.M.O Bot asks first

The bot can close a case, assign work or add a rule from chat, but every change waits for the person who asked to press Confirm.

AI YOU CONTROL

C.O.S.M.O Bot works with the AI you choose, and your data stays yours.

Ask about any page, alert or report in plain English. C.O.S.M.O Bot answers from your own data, using your own Claude Code or Codex, a government cloud your agency already approved (Azure Government, Amazon Bedrock, GSA USAi or your own AI gateway), or a model running entirely on your own hardware.



My AI Connect Claude Code or Codex, a cloud AI (Claude, OpenAI, Azure Government, Amazon Bedrock, GSA USAi or your agency's AI gateway), a model on this computer, or a model server on your network.

Anthropic	Claude Code	OpenAI · Codex	Azure Government	Amazon Bedrock	Ollama

Guardrails built in

- Text inside reports is never treated as an instruction
- Every change from chat waits for your Confirm
- An approved-model policy, and never a model from China
- Every question and tool call in the audit log

One connection for the whole team

- Connect an AI once on the server
- Every permitted account can use it
- Run fully offline with a local model
- Or switch AI off entirely; the platform still works

INTEGRATIONS

C.O.S.M.O connects to the tools you already run.

No rip and replace. C.O.S.M.O reads from your SIEM, sensors, firewalls, databases and cloud storage, read-only by design, and speaks the open standards your partners use.

SIEM and network sensors

- Splunk
- Elasticsearch
- OpenSearch
- Sentinel · Defender
- Suricata — IDS
- Zeek — NSM

Firewalls and security devices

- Palo Alto
- Fortinet
- Cisco
- Check Point — R80+
- pfSense
- AWS security groups

Databases and cloud storage

- PostgreSQL
- MySQL
- SQL Server
- Databricks
- Amazon S3
- Azure Blob

Sign-in and notifications

- Entra ID
- Okta
- Google
- Slack
- Teams
- RSS · email

Agents and automation

C.O.S.M.O also offers its tools to AI agents and scripts through MCP and a plain HTTP API, with per-key permissions and rate limits.

SECURITY AND COMPLIANCE

C.O.S.M.O is built for secure, regulated and classified environments.

A security platform is a target, so getting past one layer of C.O.S.M.O is never enough. Nothing leaves your systems unless you allow it.

Strong sign-in

- CAC and PIV smart cards
- Single sign-on (OIDC)
- Two-step verification
- Lock-out and idle sign-out

Right access for each role

- Four roles, set by a super admin
- Team duties on every account
- Integration keys capped by role
- Organizations kept apart

Evidence you can trust

- Sealed audit log with source address
- Tampering detected automatically
- Every approval named and timed
- Logs forwarded to your SIEM

Protected data

- Secrets and backups encrypted
- Keys in the system keychain
- FIPS-approved algorithm mode
- Encrypted, scheduled backups

Classified networks

- Runs fully air-gapped
- Classification banners and markings
- TLP on all intelligence
- Signed licenses and updates by file

Safe by construction

- Connectors are read-only
- Uploaded files never executed
- Feeds cannot reach internal systems
- Untrusted text cannot steer the AI

Checked against	Where it stands
SOC 2 security criteria, NIST 800-53	Controls mapped and tested in the build; ready for your auditor
OWASP Top 10 for web, API and AI apps	Each item reviewed, with tests that keep the fixes in place
DoD sign-in rules, FIPS mode	Smart cards, lock-out, idle sign-out, approved algorithms
DoDIN	Not yet listed; built ready to start the approval process

Your data stays where you put it

No cloud account needed. Offline mode loads intelligence from signed content packs, and every action is recorded with who did it and where they signed in from.

DEPLOYMENT

C.O.S.M.O runs wherever your data needs to stay.

The same software runs on one analyst's laptop, on a team server, across a large organization, or on a closed government network. Start small and grow: moving up is a backup and a restore.

Solo analyst

A desktop app for Mac, Windows or Linux with everything inside. Ideal for evaluation, a consultant's laptop or a small business.

Team server

One server the whole team opens in a browser, behind HTTPS. Accounts at every privilege level and one shared AI connection.

Large organization

Several servers sharing the load with automatic failover, read replicas and event storage spread across databases for high volume.

Air-gapped government

A closed network with smart-card sign-in, local AI on your own GPUs, classification banners, and licenses and updates delivered by file.

If you need…	Choose
To try C.O.S.M.O today	Solo analyst
A shared security operations center for a team or company	Team server
Very high event volume and high availability	Large organization
No internet, smart cards and classification markings	Air-gapped government


macOS


Windows


Linux


Docker


PostgreSQL



AWS

AT A GLANCE

Everything your security operations need ships in one platform.

No separate modules to buy and no cloud subscriptions to manage. C.O.S.M.O is one program and one database, with every capability below included.

Operations

- Eight-step guided workflow
- Incidents, cases and playbooks
- Team duties, tasks and messages
- Alert center with email and feeds
- Dashboards you can share

Intelligence

- 26 open sources built in
- STIX 2.1 knowledge base
- Reports, graphs and timelines
- Most Wanted and dark web
- Forecast of who is coming next

Detection

- Sigma, YARA, Suricata, Snort
- Beaconing and DNS tunneling
- Firewall rule analysis
- Rules proven with samples
- AI use and supplier checks
- Risky directory accounts
- Malware Lab with Ghidra

AI

- C.O.S.M.O Bot on every page
- Hosted, agent or local models
- Changes wait for your Confirm
- Prompt-injection guard
- Works without AI too

Data

- 20+ log formats
- Syslog and file sensors
- SIEM search in place
- Read-only database mirrors
- Amazon S3 and Azure storage

Governance

- Roles, SSO, MFA, smart cards
- Hash-chained audit log
- Encrypted secrets and backups
- Classification markings
- Fully offline operation

See it with your own data

A pilot runs on a single laptop or a team server, with sample traffic or your own logs, so your analysts can judge C.O.S.M.O on real work.



C.O.S.M.O

BRINGING ORDER TO CYBER OPERATIONS

To arrange a demonstration or a pilot, contact Cypher LLC.